Investigating the Cognitive Plausibility and Linguistic Interpretability of Large Language Models through Anaphora Resolution

Large Language Models (LLMs) have made remarkable progress in processing and generating human language; however, it is still an open question whether LLMs approximate human linguistic cognition **[4, 6]**, how they internally represent linguistic information **[2, 3]** and how their linguistic capabilities can be evaluated **[6]**..

This project investigates anaphora resolution, a core task in human sentence processing and a standard benchmark for assessing human-like language understanding **[5]**. It requires not only sensitivity to formal linguistic features but also the integration of world knowledge and pragmatic reasoning. In this project, we focus on a pro-drop language, namely Turkish, which has not been widely used to evaluate such language models. Our stimuli consisted of 97 items adapted for Turkish from **[1]**, and were designed to specifically probe the use of syntactic and pragmatic biases in subject pronoun interpretation towards a subject antecedent or a non-subject antecedent (see Fig 1). We tested whether LLMs responded to these biases similarly to humans.

We conducted two experiments with GPT-4o and LLaMA models (LLaMA-4 Scout 16B, LLaMA-3 8B, and Turkish-fine-tuned LLaMA-3), comparing both model outputs and attention-based internal representations with human data from native Turkish speakers (N = 23). Behavioral similarity was assessed using Pearson's correlation coefficient (ρ), while attention patterns were analyzed as a model-internal diagnostic to examine how linguistic biases are weighted within the models.

Overall, the findings presented in Fig 2 suggest that larger models, GPT-4o and LLaMA-4 were able to achieve comparable results to humans and provided more consistent responses when syntactic and pragmatic cues align, whereas human responses showed greater variability. Turkish fine-tuning helped the LLaMA-3 model show sensitivity to both syntactic and pragmatic cues while LLaMA-3 defaults to a subject antecedent (the most salient entity in discourse) across all bias conditions. None of the models replicated "someone else" responses for overt pronouns.

The attention results showed multiple candidates for "specialized" heads **[3]**, primarily in mid-to-higher layers (Fig. 3 for LLaMA-4). Rather than simply noting that specialization is distributed across several heads, we can also characterize its layer-wise profile: these candidate heads are most prominent in the mid-to-higher layers. The layer-wise analysis supports the same pattern.

Although word order was not a primary experimental manipulation, a small subset of stimuli instantiated alternative word orders within the adverbial clause. In these items, overall attention distribution appears to be highly sensitive to word order and linear proximity to target tokens, as subject and object attention ratios within most individual heads switch between SVO and OVS conditions. Moreover, despite differences in task performance, all LLaMA models exhibit a broadly similar layer-wise profile: earlier layers tend to allocate relatively more attention to the object, whereas later layers allocate relatively more attention to the subject.

Overall, these results show that LLMs vary in their sensitivity to syntactic and pragmatic biases as a function of model size and fine-tuning. Crucially, whereas humans exhibit greater response variability, the models tend to be more consistent. Attention analyses indicate that mid-to-higher layers are most predictive of model responses, and that overall attention distribution is strongly modulated by word order.

References. [1] Carminati, M. N. (2002). *The processing of Italian subject pronouns*. PhD diss. **[2]** Charpentier, F., et al. (2024). Exploring semantics in pretrained language model attention. **SEM*. **[3]** Clark, K., et al. (2019). What does BERT look at? An analysis of BERT's attention. *arXiv*. **[4]** Hu, J., et al. (2023). Pragmatic understanding in humans and language models. *ACL*. **[5]** Levesque, H. J., et al. (2012). The Winograd schema challenge. *KR*. **[6]** Mahowald, K., et al. (2024). Dissociating language and thought in large language models. *Trends Cogn. Sci.*

Conditions: Anaphor × Ambiguity (6) Q: Antecedent? (a) subject (b) object (c) someone else

Condition	Stimuli	Translation
Null, no bias	Elif Ayşe'yi gördüğünde, yemek yiyordu.	"When Elif saw Ayşe, ∅ was eating."
Null, subj bias	Elif Ayşe'yi rezil edince, özür diledi.	"When Elif humiliated Ayşe, ∅ apologized."
Null, obj bias	Elif Ayşe'yi azarlayınca, ağladı.	"When Elif scolded Ayşe, ∅ cried."
Overt, no bias	Elif Ayşe'yi gördüğünde, o yemek yiyordu.	"When Elif saw Ayşe, she was eating."
Overt, subj bias	Elif Ayşe'yi rezil edince, o özür diledi.	"When Elif humiliated Ayşe, she apologized."
Overt, obj bias	Elif Ayşe'yi azarlayınca, o ağladı	"When Elif scolded Ayşe, she cried."

Figure 1: Stimuli

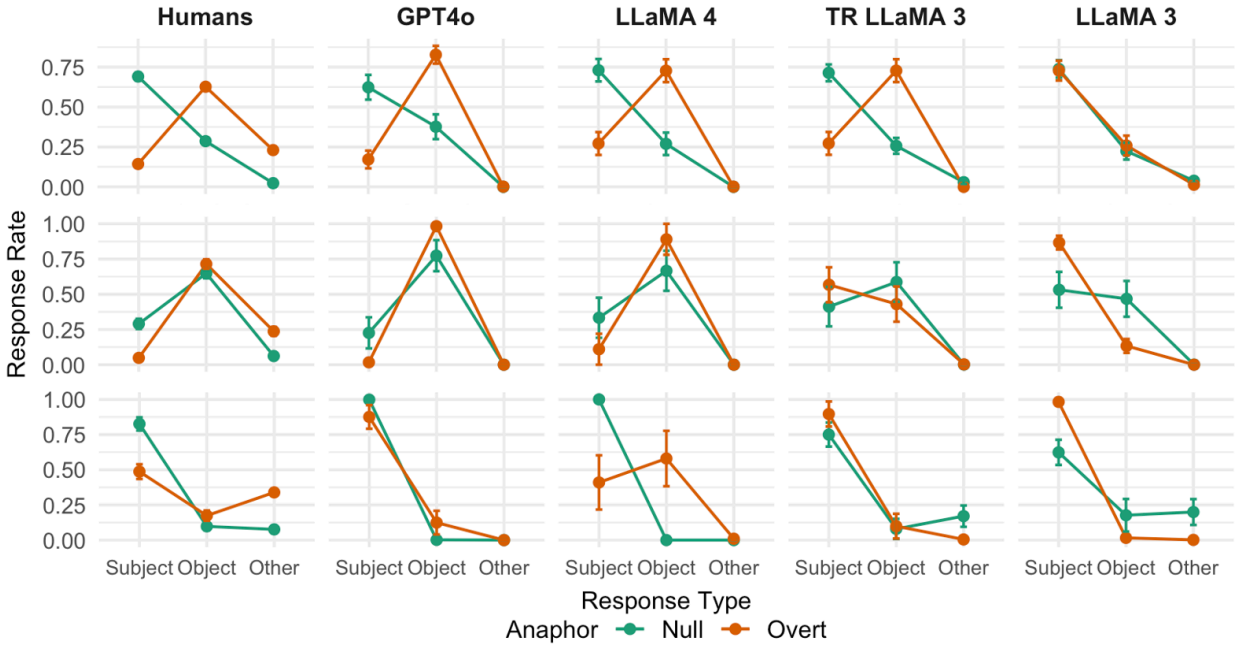

Figure 2: Response rates across models compared to human data across pragmatic bias conditions (rows) and anaphor types. Row 1=no-bias; Row 2=object-bias condition;Row 3=subject-bias.

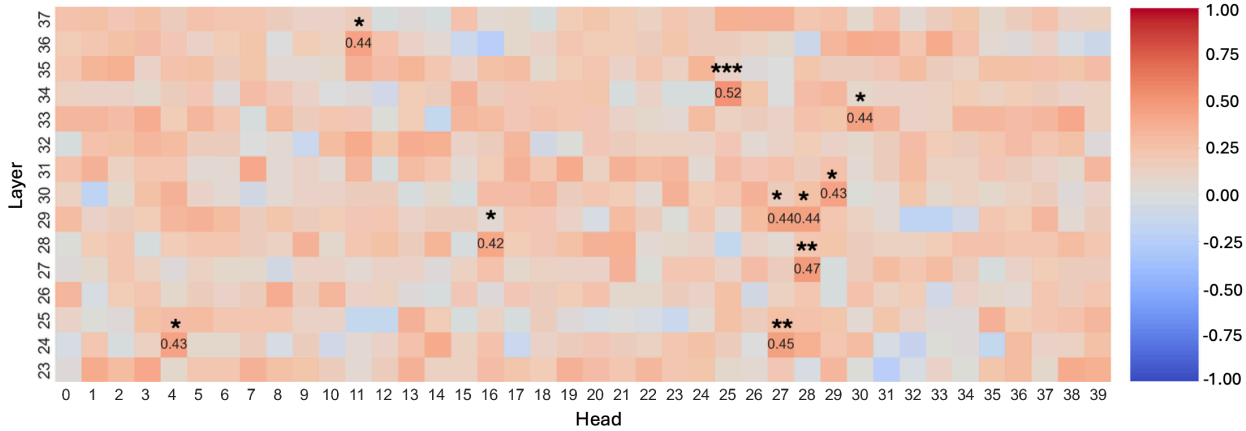

Figure 3: Headwise Pearson *r* correlations for LLaMA-4 subject-attention ratios; only layers with highly correlated results are shown. Bonferroni-corrected: *** $p < .001$, ** $p < .01$, * $p < .05$.